Enterprise Multi-Site Network Architecture Report

Inter-VLAN Routing, Static WAN Routing, Dynamic Addressing & Extended ACL Security

Author: Mohammed Moaz **Platform:** Cisco Packet Tracer **Scope:** Multi-Site Enterprise SME Network

Executive Summary & Topology Scope

This project presents the design, multi-stage implementation, and security hardening of a comprehensive Small-to-Medium Enterprise (SME) network infrastructure. The architecture transitions from a single-site segmented LAN into a dual-site WAN enterprise layout, providing dynamic addressing, isolated department domains, cross-site static routing, and fine-grained access control filtering.

Site A: Headquarters LAN

Utilizes Layer 2 VLAN segmentation across two Catalyst switches, separating **VLAN 10 (STAFF)** and **VLAN 20 (IT)** with 802.1Q trunking and Router-on-a-Stick (ROAS) sub-interfaces.

Site B: Remote Administration

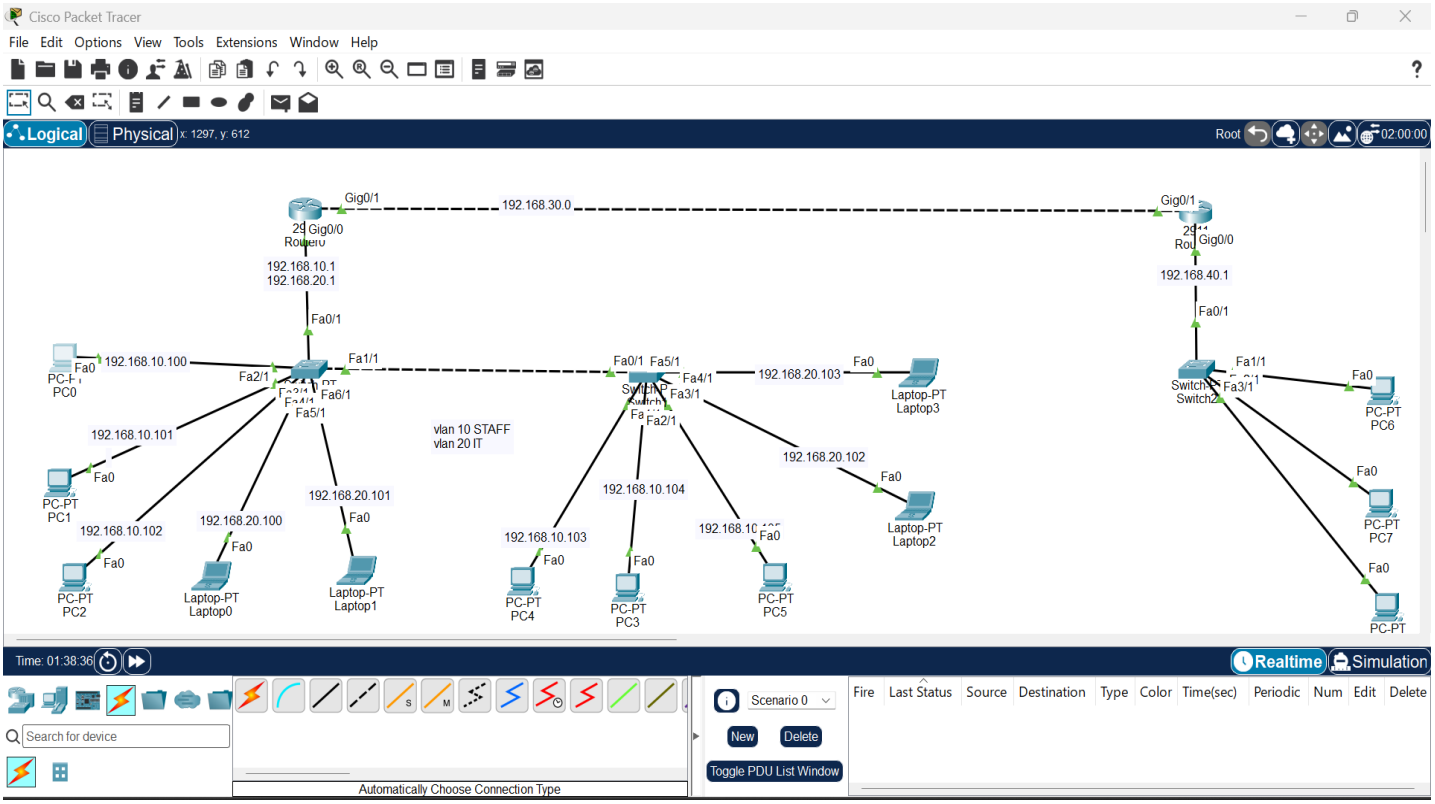
Features a standalone branch router managing an **Administration Subnet (192.168.40.0/24)** linked to HQ over a point-to-point /30 WAN interconnection.

VLAN Design & IP Address Allocation

Logical segmentation prevents broadcast storms and enforces security boundaries. Dynamic Host Configuration Protocol (DHCP) pools were deployed directly on the perimeter routers with excluded IP ranges reserved for gateways and static hosts.

VLAN / Domain	Subnet Network	Gateway Address	DHCP Pool Scope	Role / Department
VLAN 10	192.168.10.0/24	192.168.10.1	192.168.10.100 - .254	Staff Department Hosts
VLAN 20	192.168.20.0/24	192.168.20.1	192.168.20.100 - .254	IT Support & Systems
WAN Link	192.168.30.0/30	Point-to-Point	Static Allocation	Inter-Router Infrastructure
Site B Admin	192.168.40.0/24	192.168.40.1	192.168.40.100 - .254	Remote Admin Operations

1. Design



Topology Summary & Architecture Analysis:

The architecture represents a scalable, multi-site enterprise network implementation in Cisco Packet Tracer.

- **Head Office Network:** Features Layer 2 VLAN segmentation using **VLAN 10 (STAFF)** and **VLAN 20 (IT)** across interconnected switches running 802.1Q trunking. Inter-VLAN communication is handled by **Router-on-a-Stick (ROAS)** on sub-interfaces g0/0.10 (192.168.10.1/24) and g0/0.20 (192.168.20.1/24).
- **Branch Office Network:** Operates on an isolated LAN block (192.168.40.0/24) with localized router-based DHCP serving host endpoints.
- **WAN Interconnect & Routing:** The two core routers are connected over a point-to-point /30 serial/ethernet link (192.168.30.0/30). Bi-directional **Static Routes** are configured on both routers to achieve full end-to-end reachability across all site endpoints.

2. VLAN Creation and Port Assignment

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name STAFF
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface range Fa2/1,Fa3/1,Fa4/1
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface range Fa5/1,Fa6/1
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

In this step, VLAN creation and interface assignment were configured on the switch. First, VLAN 10 (named **STAFF**) and VLAN 20 (named **IT**) were created. Next, the switch interfaces were configured in access mode, where ports Fa2/1, Fa3/1, and Fa4/1 were assigned to **VLAN 10**, and ports Fa5/1 and Fa6/1 were assigned to **VLAN 20**.

3. VLAN Configuration Verification

```
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa1/1
10   STAFF                            active    Fa2/1, Fa3/1, Fa4/1
20   IT                               active    Fa5/1, Fa6/1
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

This output verifies the active VLAN configuration on the switch using the show vlan brief command. It confirms that **VLAN 10 (STAFF)** and **VLAN 20 (IT)** are active, with ports Fa2/1, Fa3/1, Fa4/1 successfully assigned to **VLAN 10** and ports Fa5/1, Fa6/1 assigned to **VLAN 20**.

4. Switch Trunk Port Configuration

```
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface Fa1/1
Switch(config-if)#sw
Switch(config-if)#switchport m
Switch(config-if)#switchport mode tr
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/1, changed state to up

Switch(config-if)#sw
Switch(config-if)#switchport tr
Switch(config-if)#switchport trunk al
Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#exit
Switch(config)#
```

a trunk port is configured on a switch interface to allow multiple VLAN traffic across a single link. First, by accessing interface Fa1/1, the port is set to trunking mode using the switchport mode trunk command. Then, the switchport trunk allowed vlan 10,20 command is executed to specify that only VLAN 10 and VLAN 20 traffic are permitted to traverse through this trunk link.

5. Router-on-a-Stick (Inter-VLAN Routing) Configuration

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface g0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#en
Router(config-subif)#encapsulation d
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface g0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
en
Router(config-subif)#encapsulation d
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
```

Inter-VLAN routing is configured on the router using the "Router-on-a-Stick" method. First, the physical interface GigabitEthernet0/0 is enabled using the no shutdown command. Then, two sub-interfaces (g0/0.10 and g0/0.20) are created. For sub-interface g0/0.10, IEEE 802.1Q encapsulation is set for VLAN 10 and assigned the IP address 192.168.10.1. Similarly, sub-interface g0/0.20 is configured with dot1Q encapsulation for VLAN 20 and assigned the IP address 192.168.20.1 to route traffic between the different VLANs

6. Router Interface Status Verification

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

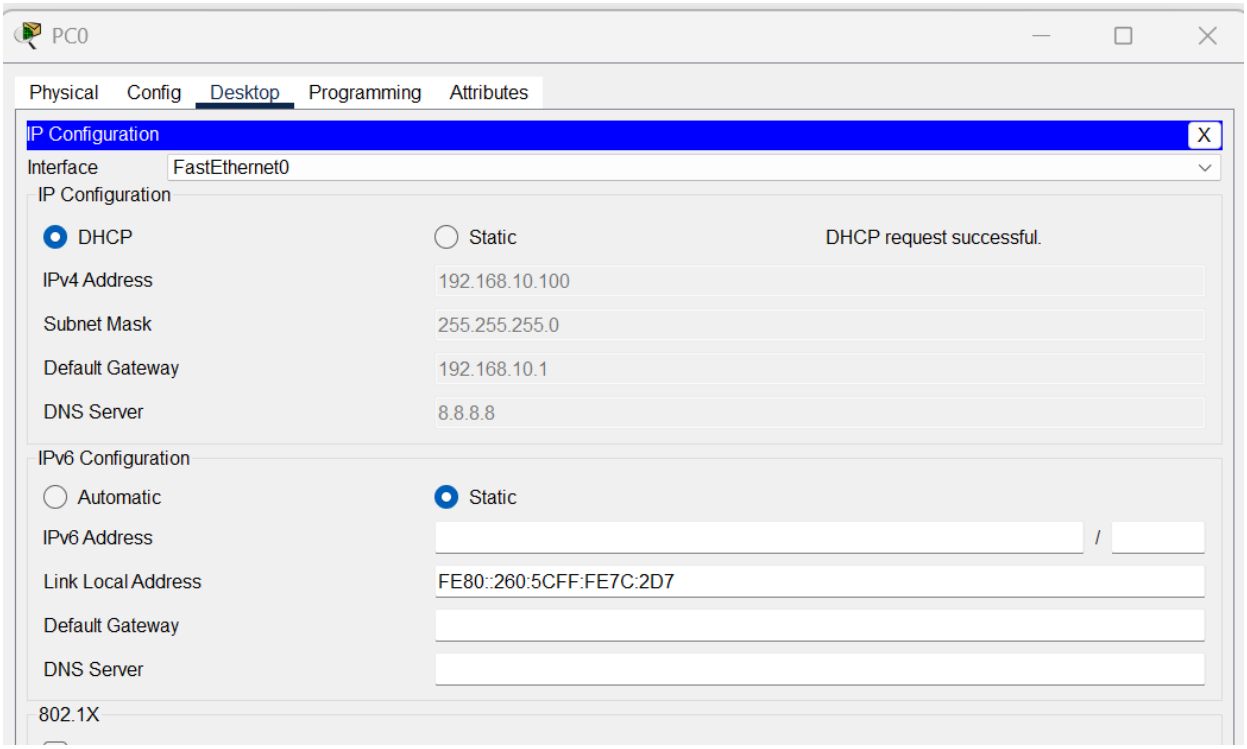
the show ip interface brief command is used on the router to verify the operational status and IP address assignments of its interfaces. The output shows that sub-interfaces GigabitEthernet0/0.10 and GigabitEthernet0/0.20 are active with IP addresses 192.168.10.1 and 192.168.20.1 respectively, and both their status and line protocol are showing as up

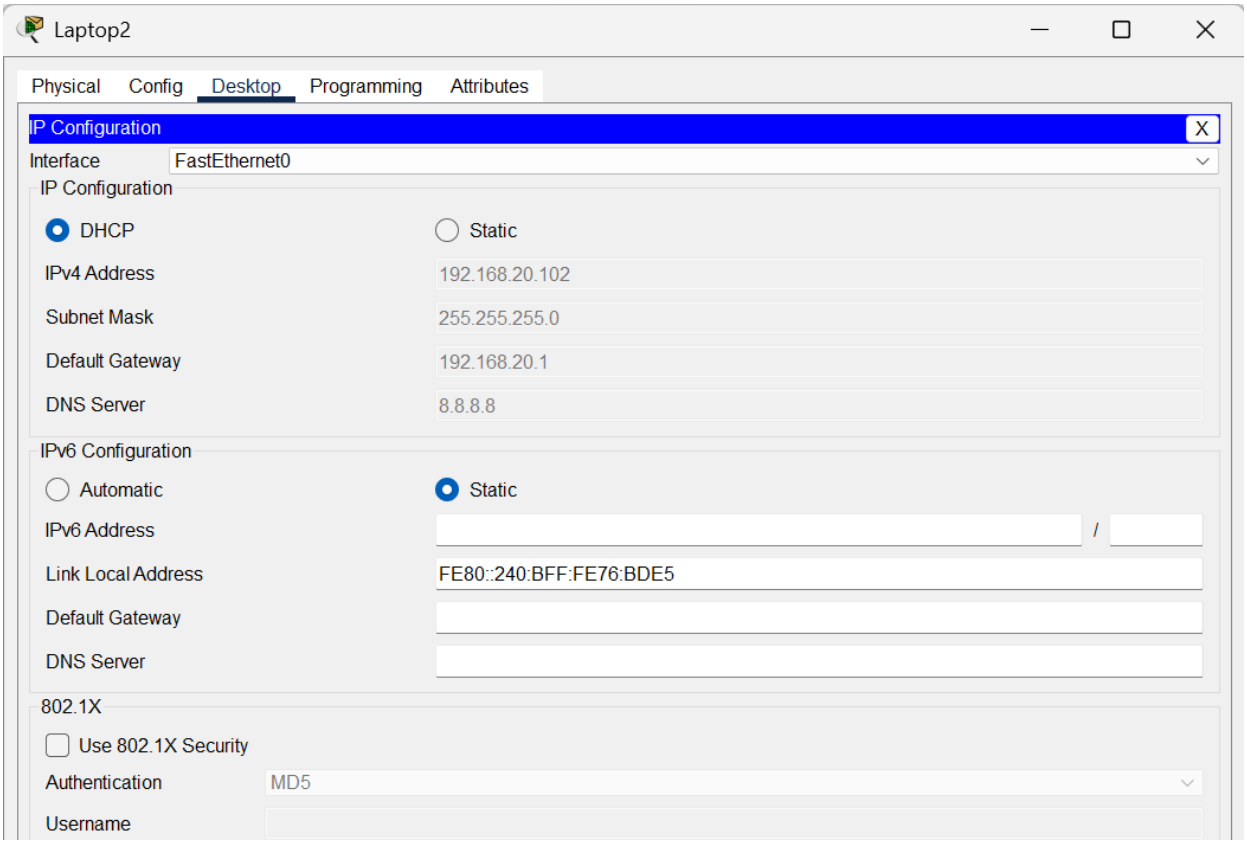
7. Router-Based DHCP Server Configuration

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp ex
Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.99
Router(config)#ip dhcp ex
Router(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.99
Router(config)#ip dhcp pool STAFF
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#def
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dn
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp pool IT
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#de
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dn
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#exit
Router#
```

In this screenshot, a router-based DHCP server is configured to automatically assign IP addresses to hosts in different VLANs. First, specific IP address ranges are excluded using the ip dhcp excluded-address command to prevent IP conflicts (ranging from 192.168.10.1 to 192.168.10.99 for the STAFF VLAN, and 192.168.20.1 to 192.168.20.99 for the IT VLAN). Then, two separate DHCP pools (STAFF and IT) are created, specifying their respective network IDs, default gateways, and the public DNS server (8.8.8.8).

8. DHCP IP Configuration Verification





9. Extended Access Control List (ACL) Configuration and Application

```
Router#
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip ac
Router(config)#ip access-list e
Router(config)#ip access-list extended STAFF-FILTER
Router(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 host 192.168.20.102
Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#exit
Router(config)#interface g0/0.10
Router(config-subif)#ip ac
Router(config-subif)#ip access-group STAFF-FILTER in
Router(config-subif)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show access-lists
Extended IP access list STAFF-FILTER
    10 deny ip 192.168.10.0 0.0.0.255 host 192.168.20.102
    20 permit ip any any

Router#
```

an Extended Access Control List named STAFF-FILTER is created and applied on the router to filter specific network traffic. An explicit rule is configured to deny any traffic coming from the 192.168.10.0/24 subnet destined for the specific host 192.168.20.102, while permitting all other IP traffic (permit ip any any). This ACL is then applied inbound on the sub-interface g0/0.10 using the ip access-group STAFF-FILTER in command, and verified using the show access-lists command.

10.ACL Enforcement Verification (ICMP Ping Test)

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: FE80::290:21FF:FE20:7C1A
   IPv6 Address.....: ::
   IPv4 Address.....: 192.168.20.101
   Subnet Mask.....: 255.255.255.0
   Default Gateway.....: ::
                                 192.168.20.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: ::
   IPv6 Address.....: ::
   IPv4 Address.....: 0.0.0.0
   Subnet Mask.....: 0.0.0.0
   Default Gateway.....: ::
                                 0.0.0.0

C:\>ping 192.168.20.102

Pinging 192.168.20.102 with 32 bytes of data:

Reply from 192.168.20.102: bytes=32 time<1ms TTL=128
Reply from 192.168.20.102: bytes=32 time<1ms TTL=128
Reply from 192.168.20.102: bytes=32 time<1ms TTL=128
Reply from 192.168.20.102: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: FE80::20C:CFFF:FE5D:EB4C
   IPv6 Address.....: ::
   IPv4 Address.....: 192.168.10.105
   Subnet Mask.....: 255.255.255.0
   Default Gateway.....: ::
                                 192.168.10.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: ::
   IPv6 Address.....: ::
   IPv4 Address.....: 0.0.0.0
   Subnet Mask.....: 0.0.0.0
   Default Gateway.....: ::
                                 0.0.0.0

C:\>ping 192.168.20.102

Pinging 192.168.20.102 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.20.102:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

These two terminal outputs demonstrate the contrast between intra-VLAN traffic flow and ACL security policy enforcement across different VLANs:

- **Intra-VLAN Communication (First Output):** The host in **VLAN 20 (IT)** with IP address 192.168.20.101 successfully pings another IT host at 192.168.20.102 with 0% packet loss. This confirms that devices within the same broadcast domain communicate without any restriction.
- **Inter-VLAN ACL Blocking (Second Output):** The host in **VLAN 10 (STAFF)** with IP address 192.168.10.105 attempts to ping the same IT host (192.168.20.102), resulting in Destination host unreachable from the default gateway 192.168.10.1. This validates that the STAFF-FILTER Access Control List on the router is actively intercepting and dropping prohibited cross-VLAN traffic while internal VLAN traffic remains unaffected.

11.Branch Router LAN Interface, DHCP Pool, and Static Route Configuration

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#interface g0/0
Router(config-if)#no shutdo
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#ip address 192.168.40.1 255.255.255.0
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#enable
Router#confi
Router#configure ter
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp pool Administration
Router(dhcp-config)#netw
Router(dhcp-config)#network 192.168.40.0 255.255.255.0
Router(dhcp-config)#de
Router(dhcp-config)#default-router 192.168.40.1
Router(dhcp-config)#dn
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#%DHCPD-4-PING_CONFLICT: DHCP address conflict:  server pinged 192.168.40.1.

Router(config)#ip route 192.168.10.0 255.255.255.0 192.168.30.1
Router(config)#ip route 192.168.20.0 255.255.255.0 192.168.30.1
Router(config)#
```

the remote/branch router is configured to support the localized site network and communicate with the main site:

- **LAN Gateway Setup:** Physical interface GigabitEthernet0/0 is brought up using no shutdown and assigned the IP address 192.168.40.1 255.255.255.0 to serve as the default gateway for the branch LAN.
- **Branch DHCP Pool:** A dedicated DHCP pool named Administration is created for the 192.168.40.0/24 subnet, configuring default gateway (192.168.40.1) and DNS server (8.8.8.8) parameters for dynamic endpoint assignment.

- **Static Routing to Head Office:** Static routes (ip route 192.168.10.0 ... and ip route 192.168.20.0 ...) are configured pointing to the next-hop IP 192.168.30.1, enabling traffic from the branch site to reach **VLAN 10** and **VLAN 20** at the head office.

12.Head Office WAN Link Interface and Static Route Configuration

```
  WCCP Redirect exclude is disabled

Router#
Router#
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)#ip address 192.168.30.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config)#ip route 192.168.40.0 255.255.255.0 192.168.30.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

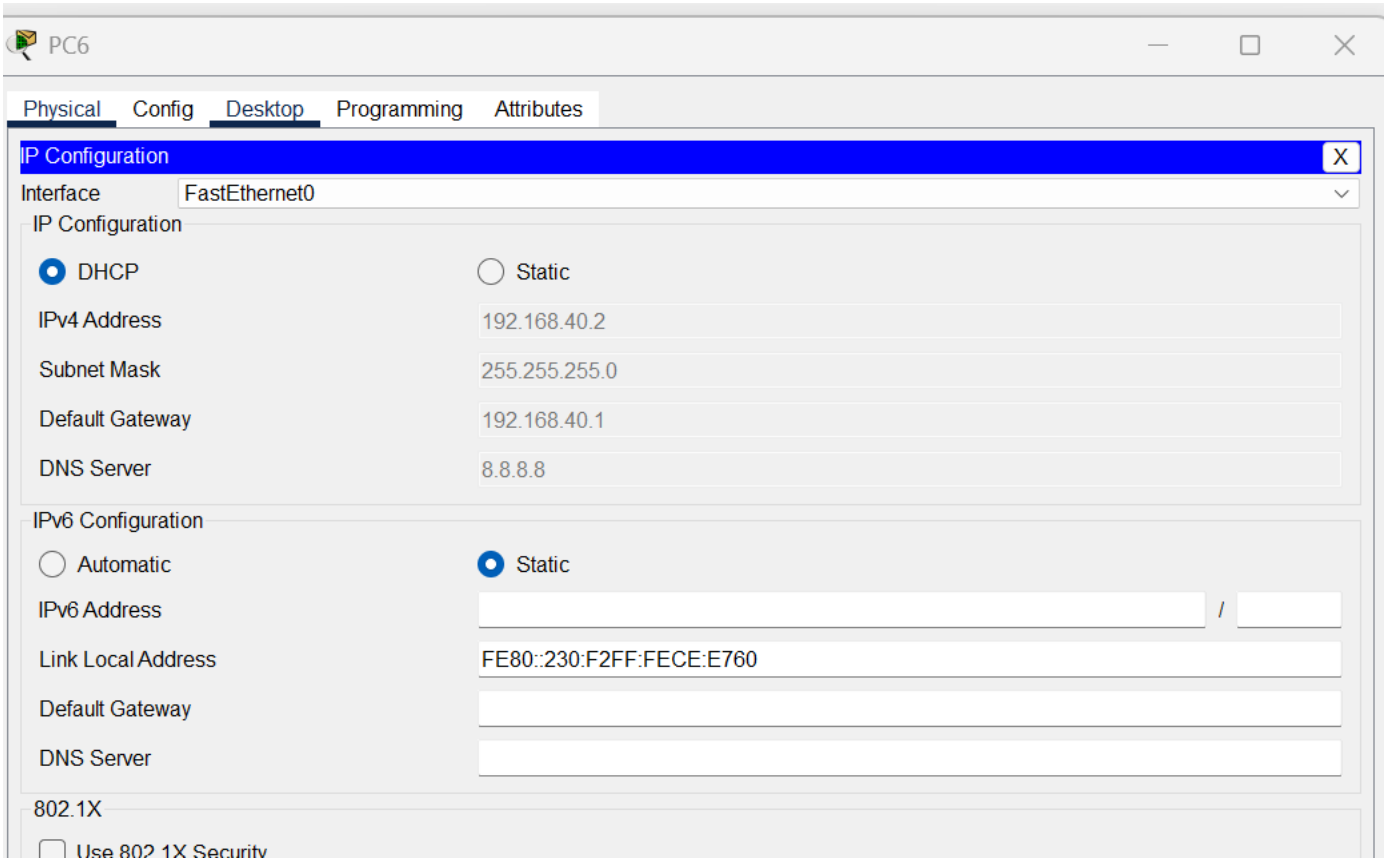
Router#
```

the head office router is configured to establish point-to-point WAN connectivity and enable routing to the remote branch site:

- **WAN Link Interface Setup:** Interface GigabitEthernet0/1 is assigned the IP address 192.168.30.1 255.255.255.252 (using a /30 point-to-point subnet mask) and brought operational via the no shutdown command.
- **Static Route Configuration:** A static route (ip route 192.168.40.0 255.255.255.0 192.168.30.2) is configured on the router. This instructs the head office router to forward all traffic destined for the remote site subnet

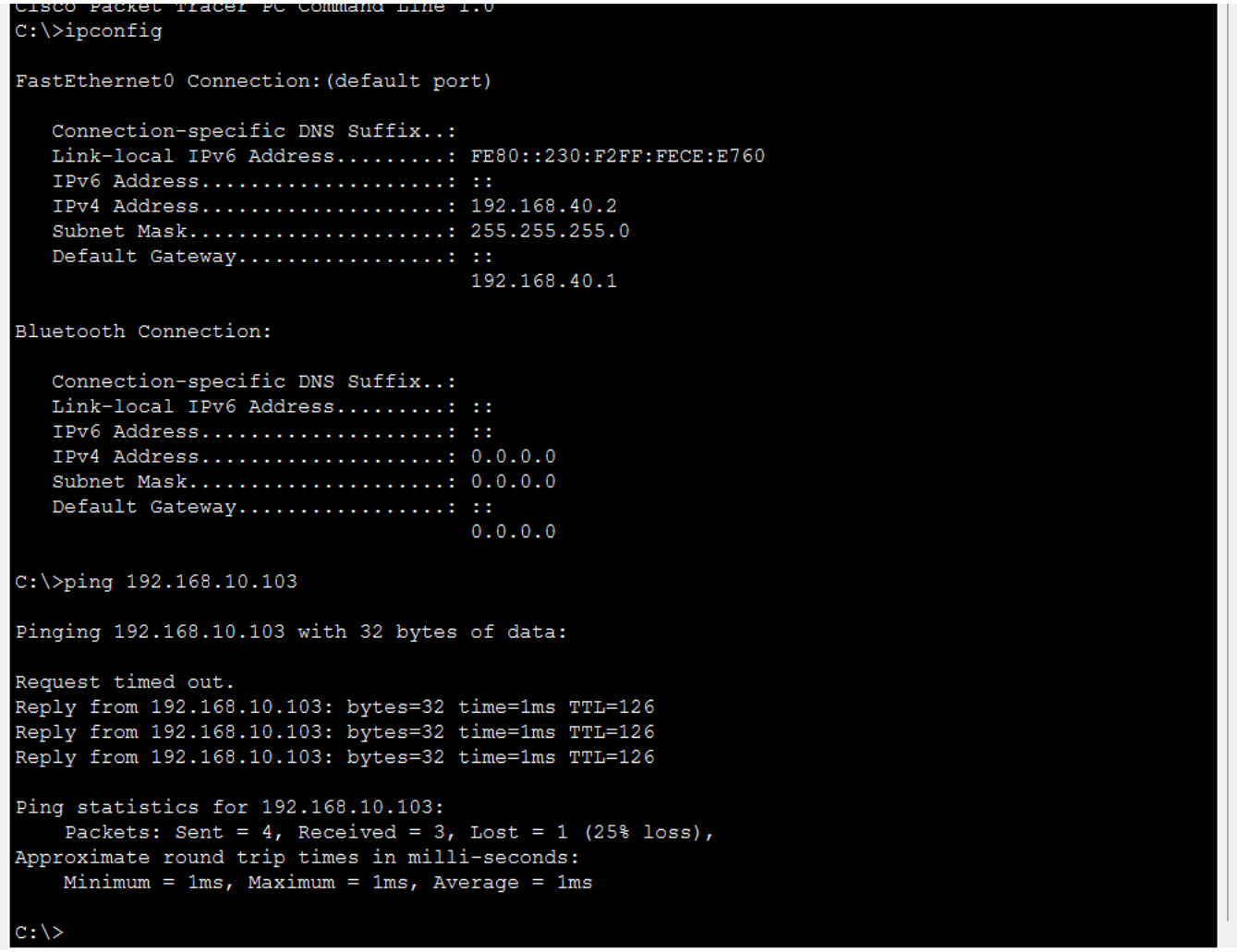
(192.168.40.0/24) to the next-hop IP address 192.168.30.2 (the branch router's WAN interface), achieving bi-directional inter-site routing.

13.Branch Site DHCP IP Configuration Verification



the IP configuration interface of **PC6** located at the branch site confirms that the host dynamically received its IP parameters from the localized Administration DHCP pool. The DHCP lease generation process completed successfully, binding IPv4 Address 192.168.40.2, Subnet Mask 255.255.255.0, Default Gateway 192.168.40.1, and DNS Server 8.8.8.8 to the host's FastEthernet0 interface

14.End-to-End Inter-Site Routing Verification (Branch PC to Head Office PC)



the ipconfig command confirms that the branch site host has an assigned IP address of 192.168.40.2 with a default gateway of 192.168.40.1. An ICMP ping command is issued to target 192.168.10.103 (a PC in VLAN 10 at the head office).

The initial packet results in a Request timed out due to the Address Resolution Protocol (ARP) table discovery process across the network, followed by three successful ICMP replies. This test verifies that static routing across the point-to-point WAN link (192.168.30.0/30) and Inter-VLAN routing at the head office are fully functional, providing complete end-to-end connectivity.

The End